

DANDI:001051: Large-scale Neuropixels recordings through SHIELD implant during visual change detection task with dynamic gating of engagement

Scope of this export (computed by this export): the units, electrodes and intervals tables of every NWB file (session files; the per-probe LFP files were skipped) were read remotely from the DANDI Archive; voltage, LFP, video and other time series were not downloaded.

Source labels: **DANDI metadata** = the Dandiset's published metadata; **NWB file** = values or text stored in the NWB files; **Computed by this export** = derived by the export scripts in `dandi_export/` (definitions given where used).

1. The dataset as published

Everything in this section is quoted from the Dandiset's published metadata or, for file-level fields, from `sub-604914_ses-20220427T041046.nwb`.

Field	Value (DANDI metadata)
Name	Large-scale Neuropixels recordings through SHIELD implant during visual change detection task with dynamic gating of engagement
Identifier / version	DANDI:001051 / 0.260922.1827
Citation	Bennett, Corbett; Sridhar, Arjun (2026) Large-scale Neuropixels recordings through SHIELD implant during visual change detection task with dynamic gating of engagement (Version 0.260922.1827) [Data set]. DANDI Archive. https://doi.org/10.48324/dandi.001051/0.260922.1827
DOI	10.48324/dandi.001051/0.260922.1827
Contributors (roles)	Bennett, Corbett (ContactPerson, Author); Sridhar, Arjun (Author); Allen Institute (Funder)
License	CC-BY-4.0
Keywords	Neuropixels, visual change detection, task engagement, SHIELD implant
Related resources	SHIELD: Skull-shaped hemispheric implants enabling large-scale electrophysiology datasets in the mouse brain doi:10.1016/j.neuron.2024.06.015 (IsDescribedBy)
Anatomy (about)	Visual areas; Striatum; visual attention; frontal cortex; medial prefrontal cortex; hippocampal formation; midbrain; Thalamus
Files / size / subjects	666 files, 1,808.07 GB, 27 subjects
Species	Mus musculus - House mouse
Approach	electrophysiological approach
Measurement technique	analytical technique; spike sorting technique; signal filtering technique; multi electrode extracellular electrophysiology recording technique
Variables measured	Units, ProcessingModule, LFP, ElectricalSeries
Date published	2026-09-22T18:27:21.645071+00:00

Description (DANDI metadata, verbatim)

This dataset was collected at the Allen Institute. It includes data from 99 electrophysiology sessions featuring multi-Neuropixel recordings throughout the left hemisphere while mice performed a visual change detection task. This dataset is closely related to the Allen Institute Visual Behavior Neuropixels (VBN) dataset, but differs in two important ways:

First, whereas the VBN recordings were focused on visual cortical areas and underlying subcortical regions, this dataset features recordings from throughout the left hemisphere, including frontal and medial cortical areas and the striatum.

Second, we have modified the behavioral task for this dataset to experimentally manipulate task engagement. During the VBN recordings, an hour of active behavior was followed by a passive behavior block during which the lick spout was retracted and mice were presented with the same visual stimuli but now with no opportunity to lick for reward. In practice, mice often satiated before the passive block. For this dataset, we interposed a 'no-reward' block in the middle of active behavior. During this new epoch, the lick spout remained extended but licks for visual changes no longer triggered reward. At the end of the no-reward block, auto-rewards were given to indicate that rewards were once again available, and many mice resumed licking for changes.

To better understand how to access and analyze this dataset, we encourage potential users to refer to the resources below. The data in DANDI is structured as follows: each subject has session NWBs identified by date of acquisition. Then, there are LFP NWBs for up to 6 probes for each session, identified by a probe id. For the LFP, each session NWB has a probes table that has the

probe ids for the LFP data associated with that session. Use this table to get the probe ids and corresponding LFP NWBs. Examples of opening a NWB file and accessing the probes table can be seen at the GitHub below under tutorials. In addition, there is a dynamic gating sessions metadata table at the GitHub repository below that has an acquisition date column, which can be used to map to a session id (the session column in the metadata tables). This will be useful for parsing the metadata tables for multi-session analysis.

- 1) This repository includes a quick-start tutorial notebook as well as metadata tables for the sessions, probes, channels and units included in this dataset:
https://github.com/AllenInstitute/SHIELD_Dynamic_Gating_Analysis
- 2) To learn more about the basic visual change detection task as well as the general structure of the nwb files, consult the documentation available for the Allen Observatory Visual Behavior Neuropixels dataset here: <https://portal.brain-map.org/circuits-behavior/visual-behavior-neuropixels>

This dataset was used in the following preprint:

SHIELD: Skull-shaped hemispheric implants enabling large-scale-electrophysiology datasets in the mouse brain [<https://doi.org/10.1101/2023.11.12.566771>]

File-level fields (NWB file, verbatim)

Field	Value (NWB file)
session_description	EPHYS_2
institution	Allen Institute for Brain Science
keywords	visual; behavior; task
subject	subject_id: 604914; species: Mus musculus; genotype: Sst-IRES-Cre/wt;Ai32(RCL-ChR2(H134R)_EYFP)/wt; sex: F; age: P203D; description: A visual behavior subject with a LabTracks ID
devices	NP.1: Allen Brain Observatory - Scientifica 2P Rig; probeA: Neuropixels 1.0 Probe; probeB: Neuropixels 1.0 Probe; probeC: Neuropixels 1.0 Probe; probeD: Neuropixels 1.0 Probe; probeE: Neuropixels 1.0 Probe; probeF: Neuropixels 1.0 Probe
electrode_groups	probeA: description = 'Ecephys Electrode Group', location = 'See electrode locations'; probeB: description = 'Ecephys Electrode Group', location = 'See electrode locations'; probeC: description = 'Ecephys Electrode Group', location = 'See electrode locations'; probeD: description = 'Ecephys Electrode Group', location = 'See electrode locations'; probeE: description = 'Ecephys Electrode Group', location = 'See electrode locations'; probeF: description = 'Ecephys Electrode Group', location = 'See electrode locations'

2. Contents of the files

Quantity	Value	Source
NWB files read / subjects	99 / 27	NWB file
Species	Mus musculus (99)	NWB file
Sex (files)	M (66), F (33)	NWB file
Genotype (files)	Sst-IRES-Cre/wt;Ai32(RCL-ChR2(H134R)_EYFP)/wt (58), wt/wt (23), Vip-IRES-Cre/wt;Ai32(RCL-ChR2(H134R)_EYFP)/wt (15), Ai32(RCL-ChR2(H134R)_EYFP)/wt (3)	NWB file
Strain (files)		NWB file
Age (files)	P172D (6), P167D (6), P225D (4), P222D (4), P208D (4), P228D (4), P180D (4), P194D (4), P205D (4), P233D (4), P234D (4), P181D (4), P185D (4), P199D (4), P196D (4), P173D (4), P186D (4), P166D (4), P183D (4), P169D (4), P165D (4), P213D (3), P195D (3), P202D (3), P203D (2)	NWB file
Lab / institution	/ Allen Institute for Brain Science (99)	NWB file
Units (total)	201,192	NWB file
Units per file	median 2039 (range 144–3086)	NWB file
Spikes (sum of spike_times_index lengths)		NWB file
How region was assigned to units	electrodes.location at units.peak_channel_id (electrode id)	Computed by this export
Distinct region labels	126	NWB file
units.quality values	good: 163,145, noise: 38,047	NWB file

The meaning of each units column, including any quality label, is quoted from the NWB file in section 7.

3. NWB files

File (session_id)	Subject	Sex	Age	Size (GB)	Units	Electrodes	Electrode groups	Interval tables (rows)
sub-604914_ses-20220427T041046	604914	F	P203D	2.01	1758	2304	probeA, probeB, probeC, probeD, probeE, probeF	dynamic_routing_image_set_presentations(5206); flash_250ms_presentations(150); gabor_20_deg_250ms_presentations(3645); spontaneous_presentations(2); trials(798)
sub-604914_ses-20220429T060155	604914	F	P203D	2.21	2262	2304	probeA, probeB, probeC, probeD, probeE, probeF	dynamic_routing_image_set_presentations(5200); flash_250ms_presentations(150); gabor_20_deg_250ms_presentations(3645); spontaneous_presentations(2); trials(487)
sub-607660_ses-20220606T223505	607660	F	P225D	2.49	2267	2304	probeA, probeB, probeC, probeD, probeE, probeF	dynamic_routing_image_set_presentations(5200); flash_250ms_presentations(150); gabor_20_deg_250ms_presentations(3645); spontaneous_presentations(2); trials(612)
sub-607660_ses-20220607T212534	607660	F	P225D	2.37	2452	2304	probeA, probeB, probeC, probeD, probeE, probeF	dynamic_routing_image_set_presentations(5198); flash_250ms_presentations(150); gabor_20_deg_250ms_presentations(3645); spontaneous_presentations(2); trials(643)
sub-607660_ses-20220608T211810	607660	F	P225D	2.54	2605	2304	probeA, probeB, probeC, probeD, probeE, probeF	dynamic_routing_image_set_presentations(5196); flash_250ms_presentations(150); gabor_20_deg_250ms_presentations(3645); spontaneous_presentations(2); trials(645)
sub-607660_ses-20220609T211116	607660	F	P225D	2.42	2375	2304	probeA, probeB, probeC, probeD, probeE, probeF	dynamic_routing_image_set_presentations(5195); flash_250ms_presentations(150); gabor_20_deg_250ms_presentations(3645); spontaneous_presentations(2); trials(645)
sub-608671_ses-20220517T214909	608671	M	P222D	1.84	1871	1920	probeA, probeC, probeD, probeE, probeF	dynamic_routing_image_set_presentations(5195); flash_250ms_presentations(150); gabor_20_deg_250ms_presentations(3645); spontaneous_presentations(2); trials(468)
sub-608671_ses-20220518T214848	608671	M	P222D	1.85	1760	2304	probeA, probeB, probeC, probeD, probeE, probeF	dynamic_routing_image_set_presentations(5204); flash_250ms_presentations(150); gabor_20_deg_250ms_presentations(3645); spontaneous_presentations(2); trials(491)
sub-608671_ses-20220519T214322	608671	M	P222D	1.93	1969	2304	probeA, probeB, probeC, probeD, probeE, probeF	dynamic_routing_image_set_presentations(5202); flash_250ms_presentations(150); gabor_20_deg_250ms_presentations(3645); spontaneous_presentations(2); trials(553)
sub-608671_ses-20220520T181402	608671	M	P222D	1.65	1278	1536	probeB, probeC, probeE, probeF	dynamic_routing_image_set_presentations(5201); flash_250ms_presentations(150); gabor_20_deg_250ms_presentations(3645); spontaneous_presentations(2); trials(639)
sub-608672_ses-20220502T221921	608672	F	P208D	2.60	2230	2304	probeA, probeB, probeC, probeD, probeE, probeF	dynamic_routing_image_set_presentations(5205); flash_250ms_presentations(150); gabor_20_deg_250ms_presentations(3645); spontaneous_presentations(2); trials(571)
sub-608672_ses-20220503T205638	608672	F	P208D	2.42	2449	2304	probeA, probeB, probeC, probeD, probeE, probeF	dynamic_routing_image_set_presentations(5198); flash_250ms_presentations(150); gabor_20_deg_250ms_presentations(3645); spontaneous_presentations(2); trials(654)
sub-608672_ses-20220504T211150	608672	F	P208D	2.61	2304	2304	probeA, probeB, probeC, probeD, probeE, probeF	dynamic_routing_image_set_presentations(5196); flash_250ms_presentations(150); gabor_20_deg_250ms_presentations(3645); spontaneous_presentations(2); trials(565)
sub-608672_ses-20220505T200416	608672	F	P208D	2.43	2315	2304	probeA, probeB, probeC, probeD, probeE, probeF	dynamic_routing_image_set_presentations(5195); flash_250ms_presentations(150); gabor_20_deg_250ms_presentations(3645); spontaneous_presentations(2); trials(638)
sub-611160_ses-20220511T220415	611160	M	P172D	2.08	2039	2304	probeA, probeB, probeC, probeD, probeE, probeF	dynamic_routing_image_set_presentations(5196); flash_250ms_presentations(150); gabor_20_deg_250ms_presentations(3645); spontaneous_presentations(2); trials(600)
sub-611160_ses-20220512T212648	611160	M	P172D	2.02	1857	2304	probeA, probeB, probeC, probeD, probeE, probeF	dynamic_routing_image_set_presentations(5201); flash_250ms_presentations(150); gabor_20_deg_250ms_presentations(3645); spontaneous_presentations(2); trials(581)

File (session_id)	Subject	Sex	Age	Size (GB)	Units	Electrodes	Electrode groups	Interval tables (rows)
sub-611160_ses-20220513T181413	611160	M	P172D	1.97	1927	1920	probeB, probeC, probeD, probeE, probeF	dynamic_routing_image_set_presentations(5204); flash_250ms_presentations(150); gabor_20_deg_250ms_presentations(3645); spontaneous_presentations(2); trials(560)
sub-611166_ses-20220705T212305	611166	F	P228D	1.92	1487	2304	probeA, probeB, probeC, probeD, probeE, probeF	dynamic_routing_image_set_presentations(5196); flash_250ms_presentations(150); gabor_20_deg_250ms_presentations(3645); spontaneous_presentations(2); trials(540)
sub-611166_ses-20220706T221604	611166	F	P228D	1.19	750	1920	probeA, probeB, probeC, probeD, probeF	dynamic_routing_image_set_presentations(5198); flash_250ms_presentations(150); gabor_20_deg_250ms_presentations(3645); spontaneous_presentations(2); trials(629)
sub-611166_ses-20220707T210213	611166	F	P228D	1.59	1011	1920	probeA, probeB, probeC, probeD, probeF	dynamic_routing_image_set_presentations(5199); flash_250ms_presentations(150); gabor_20_deg_250ms_presentations(3645); spontaneous_presentations(2); trials(629)
sub-611166_ses-20220708T172429	611166	F	P228D	1.32	921	1920	probeA, probeB, probeC, probeD, probeF	dynamic_routing_image_set_presentations(5201); flash_250ms_presentations(150); gabor_20_deg_250ms_presentations(3645); spontaneous_presentations(2); trials(563)
sub-612090_ses-20220524T212631	612090	M	P180D	3.04	3086	2304	probeA, probeB, probeC, probeD, probeE, probeF	dynamic_routing_image_set_presentations(5204); flash_250ms_presentations(150); gabor_20_deg_250ms_presentations(3645); spontaneous_presentations(2); trials(598)
sub-612090_ses-20220525T214650	612090	M	P180D	1.65	1296	1920	probeB, probeC, probeD, probeE, probeF	dynamic_routing_image_set_presentations(5201); flash_250ms_presentations(150); gabor_20_deg_250ms_presentations(3645); spontaneous_presentations(2); trials(501)
sub-612090_ses-20220526T205923	612090	M	P180D	2.60	2480	2304	probeA, probeB, probeC, probeD, probeE, probeF	dynamic_routing_image_set_presentations(5202); flash_250ms_presentations(150); gabor_20_deg_250ms_presentations(3645); spontaneous_presentations(2); trials(591)
sub-612090_ses-20220527T182739	612090	M	P180D	2.05	1867	1920	probeB, probeC, probeD, probeE, probeF	dynamic_routing_image_set_presentations(5200); flash_250ms_presentations(150); gabor_20_deg_250ms_presentations(3645); spontaneous_presentations(2); trials(516)
sub-614547_ses-20220531T221544	614547	M	P167D	2.49	2257	1920	probeA, probeB, probeD, probeE, probeF	dynamic_routing_image_set_presentations(5199); flash_250ms_presentations(150); gabor_20_deg_250ms_presentations(3645); spontaneous_presentations(2); trials(639)
sub-614547_ses-20220601T205407	614547	M	P167D	2.60	2532	2304	probeA, probeB, probeC, probeD, probeE, probeF	dynamic_routing_image_set_presentations(5199); flash_250ms_presentations(150); gabor_20_deg_250ms_presentations(3645); spontaneous_presentations(2); trials(682)
sub-614547_ses-20220603T181003	614547	M	P167D	2.06	1710	2304	probeA, probeB, probeC, probeD, probeE, probeF	dynamic_routing_image_set_presentations(5196); flash_250ms_presentations(150); gabor_20_deg_250ms_presentations(3645); spontaneous_presentations(2); trials(612)
sub-614608_ses-20220627T224458	614608	M	P194D	3.03	2853	2304	probeA, probeB, probeC, probeD, probeE, probeF	dynamic_routing_image_set_presentations(5200); flash_250ms_presentations(150); gabor_20_deg_250ms_presentations(3645); spontaneous_presentations(2); trials(542)
sub-614608_ses-20220628T211655	614608	M	P194D	2.67	2612	2304	probeA, probeB, probeC, probeD, probeE, probeF	dynamic_routing_image_set_presentations(5204); flash_250ms_presentations(150); gabor_20_deg_250ms_presentations(3645); spontaneous_presentations(2); trials(516)
sub-614608_ses-20220629T220553	614608	M	P194D	2.60	2356	1920	probeB, probeC, probeD, probeE, probeF	dynamic_routing_image_set_presentations(5197); flash_250ms_presentations(150); gabor_20_deg_250ms_presentations(3645); spontaneous_presentations(2); trials(550)
sub-614608_ses-20220630T213929	614608	M	P194D	2.38	1988	2304	probeA, probeB, probeC, probeD, probeE, probeF	dynamic_routing_image_set_presentations(5198); flash_250ms_presentations(150); gabor_20_deg_250ms_presentations(3645); spontaneous_presentations(2); trials(515)
sub-615047_ses-20220711T224005	615047	M	P205D	2.36	2358	2304	probeA, probeB, probeC, probeD, probeE, probeF	dynamic_routing_image_set_presentations(5204); flash_250ms_presentations(150); gabor_20_deg_250ms_presentations(3645); spontaneous_presentations(2); trials(513)

File (session_id)	Subject	Sex	Age	Size (GB)	Units	Electrodes	Electrode groups	Interval tables (rows)
sub-615047_ses-20220712T211809	615047	M	P205D	2.09	1914	2304	probeA, probeB, probeC, probeD, probeE, probeF	dynamic_routing_image_set_presentations(5205); flash_250ms_presentations(150); gabor_20_deg_250ms_presentations(3645); spontaneous_presentations(2); trials(508)
sub-615047_ses-20220713T211752	615047	M	P205D	1.95	1740	2304	probeA, probeB, probeC, probeD, probeE, probeF	dynamic_routing_image_set_presentations(5196); flash_250ms_presentations(150); gabor_20_deg_250ms_presentations(3645); spontaneous_presentations(2); trials(804)
sub-615047_ses-20220714T203802	615047	M	P205D	1.94	1699	2304	probeA, probeB, probeC, probeD, probeE, probeF	dynamic_routing_image_set_presentations(5196); flash_250ms_presentations(150); gabor_20_deg_250ms_presentations(3645); spontaneous_presentations(2); trials(858)
sub-615048_ses-20220809T213337	615048	M	P233D	2.07	2559	2304	probeA, probeB, probeC, probeD, probeE, probeF	flash_250ms_presentations(150); gabor_20_deg_250ms_presentations(3645); image-set-im104_r_presentations(5199); spontaneous_presentations(2); trials(676)
sub-615048_ses-20220810T211453	615048	M	P233D	2.08	2246	2304	probeA, probeB, probeC, probeD, probeE, probeF	flash_250ms_presentations(150); gabor_20_deg_250ms_presentations(3645); image-set-im114_r_presentations(5196); spontaneous_presentations(2); trials(675)
sub-615048_ses-20220811T211342	615048	M	P233D	2.16	2742	2304	probeA, probeB, probeC, probeD, probeE, probeF	flash_250ms_presentations(150); gabor_20_deg_250ms_presentations(3645); image-set-im005_r_presentations(5196); spontaneous_presentations(2); trials(599)
sub-615048_ses-20220812T162115	615048	M	P233D	2.06	2557	2304	probeA, probeB, probeC, probeD, probeE, probeF	flash_250ms_presentations(150); gabor_20_deg_250ms_presentations(3645); image-set-im087_r_presentations(5197); spontaneous_presentations(2); trials(600)
sub-615563_ses-20220726T223606	615563	M	P213D	2.32	2878	2304	probeA, probeB, probeC, probeD, probeE, probeF	flash_250ms_presentations(150); gabor_20_deg_250ms_presentations(3645); image-set-im104_r_presentations(5196); spontaneous_presentations(2); trials(677)
sub-615563_ses-20220728T204852	615563	M	P213D	1.72	2106	2304	probeA, probeB, probeC, probeD, probeE, probeF	flash_250ms_presentations(150); gabor_20_deg_250ms_presentations(3645); image-set-im005_r_presentations(5201); spontaneous_presentations(2); trials(611)
sub-615563_ses-20220729T172702	615563	M	P213D	1.59	1975	2304	probeA, probeB, probeC, probeD, probeE, probeF	flash_250ms_presentations(150); gabor_20_deg_250ms_presentations(3645); image-set-im087_r_presentations(5197); spontaneous_presentations(2); trials(691)
sub-615564_ses-20220815T222607	615564	M	P234D	2.18	2460	2304	probeA, probeB, probeC, probeD, probeE, probeF	flash_250ms_presentations(150); gabor_20_deg_250ms_presentations(3645); image-set-im104_r_presentations(5196); spontaneous_presentations(2); trials(921)
sub-615564_ses-20220816T201007	615564	M	P234D	2.44	2604	2304	probeA, probeB, probeC, probeD, probeE, probeF	flash_250ms_presentations(150); gabor_20_deg_250ms_presentations(3645); image-set-im114_r_presentations(5206); spontaneous_presentations(2); trials(747)
sub-615564_ses-20220817T213756	615564	M	P234D	1.76	2074	2304	probeA, probeB, probeC, probeD, probeE, probeF	flash_250ms_presentations(150); gabor_20_deg_250ms_presentations(3645); image-set-im005_r_presentations(5200); spontaneous_presentations(2); trials(705)
sub-615564_ses-20220818T210203	615564	M	P234D	1.83	2078	2304	probeA, probeB, probeC, probeD, probeE, probeF	flash_250ms_presentations(150); gabor_20_deg_250ms_presentations(3645); image-set-im087_r_presentations(5195); spontaneous_presentations(2); trials(903)
sub-623319_ses-20220907T210904	623319	M	P195D	1.65	1948	2304	probeA, probeB, probeC, probeD, probeE, probeF	flash_250ms_presentations(150); gabor_20_deg_250ms_presentations(3645); image-set-im114_r_presentations(5199); spontaneous_presentations(2); trials(752)
sub-623319_ses-20220908T212131	623319	M	P195D	1.76	1862	2304	probeA, probeB, probeC, probeD, probeE, probeF	flash_250ms_presentations(150); gabor_20_deg_250ms_presentations(3645); image-set-im005_r_presentations(5198); spontaneous_presentations(2); trials(755)
sub-623319_ses-20220909T175625	623319	M	P195D	1.83	2408	2304	probeA, probeB, probeC, probeD, probeE, probeF	flash_250ms_presentations(150); gabor_20_deg_250ms_presentations(3645); image-set-im087_r_presentations(5201); spontaneous_presentations(2); trials(679)

File (session_id)	Subject	Sex	Age	Size (GB)	Units	Electrodes	Electrode groups	Interval tables (rows)
sub-623322_ses-20220822T213256	623322	M	P181D	1.94	1921	2304	probeA, probeB, probeC, probeD, probeE, probeF	flash_250ms_presentations(150); gabor_20_deg_250ms_presentations(3645); image-set-im104_r_presentations(5201); spontaneous_presentations(2); trials(722)
sub-623322_ses-20220823T222346	623322	M	P181D	1.73	1946	2304	probeA, probeB, probeC, probeD, probeE, probeF	flash_250ms_presentations(150); gabor_20_deg_250ms_presentations(3645); image-set-im114_r_presentations(5199); spontaneous_presentations(2); trials(806)
sub-623322_ses-20220824T212827	623322	M	P181D	1.97	1954	2304	probeA, probeB, probeC, probeD, probeE, probeF	flash_250ms_presentations(150); gabor_20_deg_250ms_presentations(3645); image-set-im005_r_presentations(5198); spontaneous_presentations(2); trials(652)
sub-623322_ses-20220825T212029	623322	M	P181D	1.10	1854	2304	probeA, probeB, probeC, probeD, probeE, probeF	flash_250ms_presentations(150); gabor_20_deg_250ms_presentations(3645); image-set-im087_r_presentations(5199); spontaneous_presentations(2); trials(727)
sub-623784_ses-20220829T215528	623784	M	P185D	1.65	1575	2304	probeA, probeB, probeC, probeD, probeE, probeF	flash_250ms_presentations(150); gabor_20_deg_250ms_presentations(3645); image-set-im104_r_presentations(5200); spontaneous_presentations(2); trials(1001)
sub-623784_ses-20220830T212933	623784	M	P185D	1.80	2130	2304	probeA, probeB, probeC, probeD, probeE, probeF	flash_250ms_presentations(150); gabor_20_deg_250ms_presentations(3645); image-set-im114_r_presentations(5198); spontaneous_presentations(2); trials(949)
sub-623784_ses-20220831T212027	623784	M	P185D	1.28	1326	2304	probeA, probeB, probeC, probeD, probeE, probeF	flash_250ms_presentations(150); gabor_20_deg_250ms_presentations(3645); image-set-im005_r_presentations(5196); spontaneous_presentations(2); trials(845)
sub-623784_ses-20220901T200844	623784	M	P185D	1.68	1975	2304	probeA, probeB, probeC, probeD, probeE, probeF	flash_250ms_presentations(150); gabor_20_deg_250ms_presentations(3645); image-set-im087_r_presentations(5195); spontaneous_presentations(2); trials(1028)
sub-623786_ses-20220912T220716	623786	M	P199D	1.96	2079	2304	probeA, probeB, probeC, probeD, probeE, probeF	flash_250ms_presentations(150); gabor_20_deg_250ms_presentations(3645); image-set-im104_r_presentations(5197); spontaneous_presentations(2); trials(1182)
sub-623786_ses-20220913T211940	623786	M	P199D	1.67	1858	2304	probeA, probeB, probeC, probeD, probeE, probeF	flash_250ms_presentations(150); gabor_20_deg_250ms_presentations(3645); image-set-im114_r_presentations(5200); spontaneous_presentations(2); trials(732)
sub-623786_ses-20220914T210304	623786	M	P199D	2.06	2469	2304	probeA, probeB, probeC, probeD, probeE, probeF	flash_250ms_presentations(150); gabor_20_deg_250ms_presentations(3645); image-set-im005_r_presentations(5199); spontaneous_presentations(2); trials(701)
sub-623786_ses-20220915T202517	623786	M	P199D	1.96	2351	2304	probeA, probeB, probeC, probeD, probeE, probeF	flash_250ms_presentations(150); gabor_20_deg_250ms_presentations(3645); image-set-im087_r_presentations(5201); spontaneous_presentations(2); trials(670)
sub-626279_ses-20220926T212457	626279	M	P196D	2.18	2192	2304	probeA, probeB, probeC, probeD, probeE, probeF	flash_250ms_presentations(150); gabor_20_deg_250ms_presentations(3645); image-set-im104_r_presentations(5196); spontaneous_presentations(2); trials(751)
sub-626279_ses-20220927T215524	626279	M	P196D	0.30	144	384	probeA	flash_250ms_presentations(150); gabor_20_deg_250ms_presentations(3645); image-set-im114_r_presentations(5205); spontaneous_presentations(2); trials(573)
sub-626279_ses-20220928T210634	626279	M	P196D	2.04	2125	2304	probeA, probeB, probeC, probeD, probeE, probeF	flash_250ms_presentations(150); gabor_20_deg_250ms_presentations(3645); image-set-im005_r_presentations(5201); spontaneous_presentations(2); trials(632)
sub-626279_ses-20220929T205157	626279	M	P196D	1.47	1517	2304	probeA, probeB, probeC, probeD, probeE, probeF	flash_250ms_presentations(150); gabor_20_deg_250ms_presentations(3645); image-set-im087_r_presentations(5196); spontaneous_presentations(2); trials(714)
sub-632295_ses-20221017T220100	632295	F	P173D	2.12	2212	2304	probeA, probeB, probeC, probeD, probeE, probeF	flash_250ms_presentations(150); gabor_20_deg_250ms_presentations(3645); image-set-im104_r_presentations(5195); spontaneous_presentations(2); trials(793)

File (session_id)	Subject	Sex	Age	Size (GB)	Units	Electrodes	Electrode groups	Interval tables (rows)
sub-632295_ses-20221018T215811	632295	F	P173D	2.04	2418	2304	probeA, probeB, probeC, probeD, probeE, probeF	flash_250ms_presentations(150); gabor_20_deg_250ms_presentations(3645); image-set-im114_r_presentations(5196); spontaneous_presentations(2); trials(792)
sub-632295_ses-20221019T215724	632295	F	P173D	1.78	1719	2304	probeA, probeB, probeC, probeD, probeE, probeF	flash_250ms_presentations(150); gabor_20_deg_250ms_presentations(3645); image-set-im005_r_presentations(5203); spontaneous_presentations(2); trials(721)
sub-632295_ses-20221020T215833	632295	F	P173D	1.85	2029	2304	probeA, probeB, probeC, probeD, probeE, probeF	flash_250ms_presentations(150); gabor_20_deg_250ms_presentations(3645); image-set-im087_r_presentations(5203); spontaneous_presentations(2); trials(707)
sub-633232_ses-20221108T000611	633232	F	P186D	1.00	1088	2304	probeA, probeB, probeC, probeD, probeE, probeF	flash_250ms_presentations(150); gabor_20_deg_250ms_presentations(3645); image-set-im104_r_presentations(5208); spontaneous_presentations(2); trials(963)
sub-633232_ses-20221108T230110	633232	F	P186D	1.71	1933	2304	probeA, probeB, probeC, probeD, probeE, probeF	flash_250ms_presentations(150); gabor_20_deg_250ms_presentations(3645); image-set-im114_r_presentations(5203); spontaneous_presentations(2); trials(590)
sub-633232_ses-20221109T225948	633232	F	P186D	1.27	1451	2304	probeA, probeB, probeC, probeD, probeE, probeF	flash_250ms_presentations(150); gabor_20_deg_250ms_presentations(3645); image-set-im005_r_presentations(5204); spontaneous_presentations(2); trials(728)
sub-633232_ses-20221110T225606	633232	F	P186D	1.37	1461	2304	probeA, probeB, probeC, probeD, probeE, probeF	flash_250ms_presentations(150); gabor_20_deg_250ms_presentations(3645); image-set-im087_r_presentations(5199); spontaneous_presentations(2); trials(821)
sub-636740_ses-20221115T052954	636740	F	P166D	1.54	1938	2304	probeA, probeB, probeC, probeD, probeE, probeF	flash_250ms_presentations(150); gabor_20_deg_250ms_presentations(3645); image-set-im104_r_presentations(5200); spontaneous_presentations(2); trials(472)
sub-636740_ses-20221115T214032	636740	F	P166D	2.29	2665	2304	probeA, probeB, probeC, probeD, probeE, probeF	flash_250ms_presentations(150); gabor_20_deg_250ms_presentations(3645); image-set-im114_r_presentations(5200); spontaneous_presentations(2); trials(555)
sub-636740_ses-20221116T225638	636740	F	P166D	1.69	1629	2304	probeA, probeB, probeC, probeD, probeE, probeF	flash_250ms_presentations(150); gabor_20_deg_250ms_presentations(3645); image-set-im005_r_presentations(5202); spontaneous_presentations(2); trials(569)
sub-636740_ses-20221117T230548	636740	F	P166D	1.43	2402	2304	probeA, probeB, probeC, probeD, probeE, probeF	flash_250ms_presentations(150); gabor_20_deg_250ms_presentations(3645); image-set-im087_r_presentations(5196); spontaneous_presentations(2); trials(620)
sub-637483_ses-20221121T225619	637483	M	P167D	1.76	1815	2304	probeA, probeB, probeC, probeD, probeE, probeF	flash_250ms_presentations(150); gabor_20_deg_250ms_presentations(3645); image-set-im104_r_presentations(5200); spontaneous_presentations(2); trials(790)
sub-637483_ses-20221122T232820	637483	M	P167D	2.35	2865	2304	probeA, probeB, probeC, probeD, probeE, probeF	flash_250ms_presentations(150); gabor_20_deg_250ms_presentations(3645); image-set-im114_r_presentations(5200); spontaneous_presentations(2); trials(697)
sub-637483_ses-20221123T172329	637483	M	P167D	1.44	1817	2304	probeA, probeB, probeC, probeD, probeE, probeF	flash_250ms_presentations(150); gabor_20_deg_250ms_presentations(3645); image-set-im005_r_presentations(5204); spontaneous_presentations(2); trials(727)
sub-637488_ses-20221206T084803	637488	F	P183D	1.56	2261	2304	probeA, probeB, probeC, probeD, probeE, probeF	flash_250ms_presentations(150); gabor_20_deg_250ms_presentations(3645); image-set-im104_r_presentations(5205); spontaneous_presentations(2); trials(439)
sub-637488_ses-20221207T082121	637488	F	P183D	1.70	2328	2304	probeA, probeB, probeC, probeD, probeE, probeF	flash_250ms_presentations(150); gabor_20_deg_250ms_presentations(3645); image-set-im114_r_presentations(5197); spontaneous_presentations(2); trials(452)
sub-637488_ses-20221208T063534	637488	F	P183D	1.82	2573	2304	probeA, probeB, probeC, probeD, probeE, probeF	flash_250ms_presentations(150); gabor_20_deg_250ms_presentations(3645); image-set-im005_r_presentations(5205); spontaneous_presentations(2); trials(526)

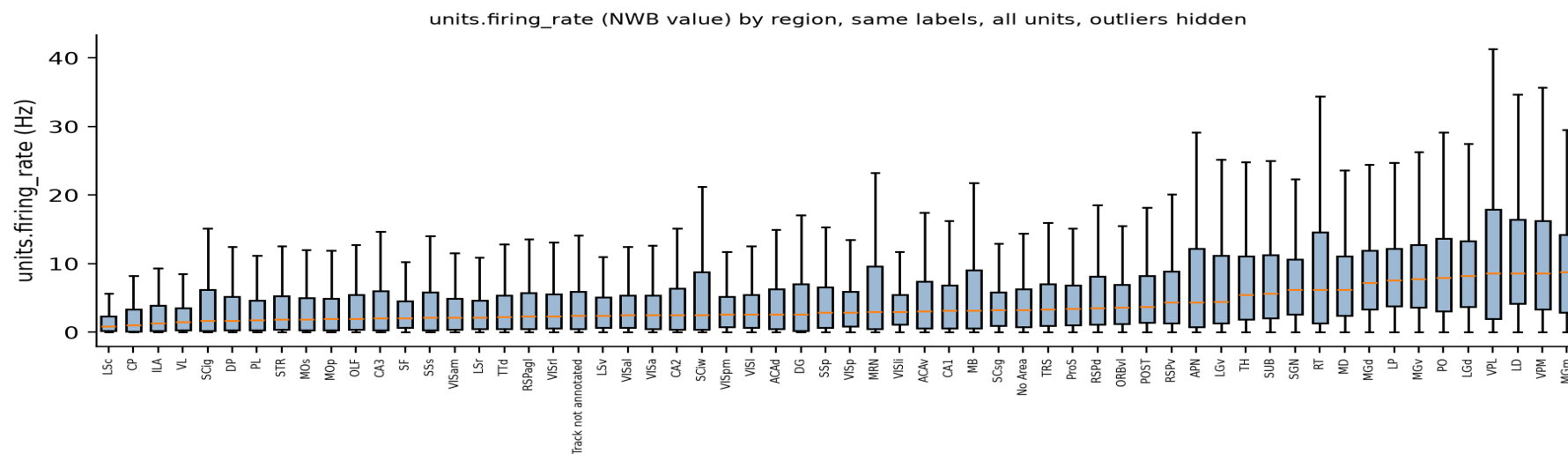
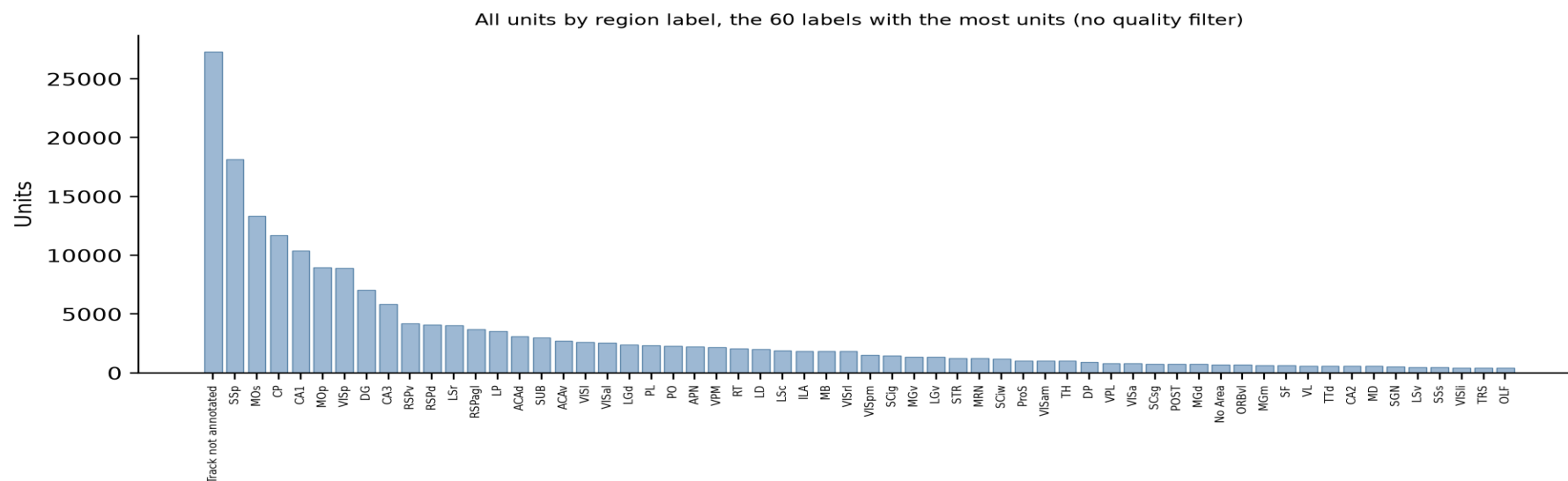
File (session_id)	Subject	Sex	Age	Size (GB)	Units	Electrodes	Electrode groups	Interval tables (rows)
sub-637488_ses-20221209T060640	637488	F	P183D	1.96	2500	2304	probeA, probeB, probeC, probeD, probeE, probeF	flash_250ms_presentations(150); gabor_20_deg_250ms_presentations(3645); image-set-im087_r_presentations(5198); spontaneous_presentations(2); trials(495)
sub-638387_ses-20221130T063555	638387	M	P169D	2.25	2453	2304	probeA, probeB, probeC, probeD, probeE, probeF	flash_250ms_presentations(150); gabor_20_deg_250ms_presentations(3645); image-set-im104_r_presentations(5208); spontaneous_presentations(2); trials(628)
sub-638387_ses-20221201T063548	638387	M	P169D	1.70	1861	2304	probeA, probeB, probeC, probeD, probeE, probeF	flash_250ms_presentations(150); gabor_20_deg_250ms_presentations(3645); image-set-im114_r_presentations(5197); spontaneous_presentations(2); trials(1101)
sub-638387_ses-20221202T063256	638387	M	P169D	2.23	2397	2304	probeA, probeB, probeC, probeD, probeE, probeF	flash_250ms_presentations(150); gabor_20_deg_250ms_presentations(3645); image-set-im005_r_presentations(5205); spontaneous_presentations(2); trials(834)
sub-638387_ses-20221203T032412	638387	M	P169D	1.05	1302	2304	probeA, probeB, probeC, probeD, probeE, probeF	flash_250ms_presentations(150); gabor_20_deg_250ms_presentations(3645); image-set-im087_r_presentations(5198); spontaneous_presentations(2); trials(954)
sub-640887_ses-20221220T073312	640887	M	P172D	1.85	2022	2304	probeA, probeB, probeC, probeD, probeE, probeF	flash_250ms_presentations(150); gabor_20_deg_250ms_presentations(3645); image-set-im104_r_presentations(5195); spontaneous_presentations(2); trials(596)
sub-640887_ses-20221221T035211	640887	M	P172D	1.71	2082	2304	probeA, probeB, probeC, probeD, probeE, probeF	flash_250ms_presentations(150); gabor_20_deg_250ms_presentations(3645); image-set-im114_r_presentations(5200); spontaneous_presentations(2); trials(557)
sub-640887_ses-20221222T040053	640887	M	P172D	1.92	2096	2304	probeA, probeB, probeC, probeD, probeE, probeF	flash_250ms_presentations(150); gabor_20_deg_250ms_presentations(3645); image-set-im005_r_presentations(5202); spontaneous_presentations(2); trials(544)
sub-640890_ses-20221213T065302	640890	M	P165D	2.29	2667	2304	probeA, probeB, probeC, probeD, probeE, probeF	flash_250ms_presentations(150); gabor_20_deg_250ms_presentations(3645); image-set-im104_r_presentations(5198); spontaneous_presentations(2); trials(570)
sub-640890_ses-20221214T070918	640890	M	P165D	1.65	1959	2304	probeA, probeB, probeC, probeD, probeE, probeF	flash_250ms_presentations(150); gabor_20_deg_250ms_presentations(3645); image-set-im114_r_presentations(5207); spontaneous_presentations(2); trials(567)
sub-640890_ses-20221215T053357	640890	M	P165D	1.99	2315	2304	probeA, probeB, probeC, probeD, probeE, probeF	flash_250ms_presentations(150); gabor_20_deg_250ms_presentations(3645); image-set-im005_r_presentations(5199); spontaneous_presentations(2); trials(627)
sub-640890_ses-20221216T063330	640890	M	P165D	1.45	1678	2304	probeA, probeB, probeC, probeD, probeE, probeF	flash_250ms_presentations(150); gabor_20_deg_250ms_presentations(3645); image-set-im087_r_presentations(5197); spontaneous_presentations(2); trials(615)
sub-642504_ses-20230131T073942	642504	F	P202D	1.19	1419	2304	probeA, probeB, probeC, probeD, probeE, probeF	flash_250ms_presentations(150); gabor_20_deg_250ms_presentations(3645); image-set-im104_r_presentations(5200); spontaneous_presentations(2); trials(556)
sub-642504_ses-20230201T065423	642504	F	P202D	1.55	1607	2304	probeA, probeB, probeC, probeD, probeE, probeF	flash_250ms_presentations(150); gabor_20_deg_250ms_presentations(3645); image-set-im114_r_presentations(5197); spontaneous_presentations(2); trials(772)
sub-642504_ses-20230202T071036	642504	F	P202D	1.51	1682	2304	probeA, probeB, probeC, probeD, probeE, probeF	flash_250ms_presentations(150); gabor_20_deg_250ms_presentations(3645); image-set-im005_r_presentations(5208); spontaneous_presentations(2); trials(620)

Other data in each file (not exported)

File (session_id)	Acquisition	Processing modules: interfaces
sub-604914_ses-20220427T041046	EyeTracking, v_in, v_sig	eye_tracking_rig_metadata:eye_tracking_rig_metadata; licking:licks; optotagging:optotagging/optogenetic_stimulation; rewards:autorewarded/volume; running:dx/speed/speed_unfiltered; stimulus:timestamps

File (session_id)	Acquisition	Processing modules: interfaces
sub-642504_ses-20230201T065423	EyeTracking, v_in, v_sig	eye_tracking_rig_metadata:eye_tracking_rig_metadata; licking:licks; optotagging:optotagging/optogenetic_stimulation; rewards:autorewarded/volume; running:dx/speed/speed_unfiltered; stimulus:timestamps
sub-642504_ses-20230202T071036	EyeTracking, v_in, v_sig	eye_tracking_rig_metadata:eye_tracking_rig_metadata; licking:licks; optotagging:optotagging/optogenetic_stimulation; rewards:autorewarded/volume; running:dx/speed/speed_unfiltered; stimulus:timestamps

4. Units per region (Computed by this export)



5. All region labels (Computed by this export)

One row per region label exactly as stored in the NWB files, all units (no quality filter), sorted by unit count. Firing-rate columns summarise the NWB units.firing_rate values where the dataset provides them.

Region label	Units	Files	Subjects	firing_rate mean (Hz)	firing_rate median (Hz)
Track not annotated	27301	45	19	4.63	2.40
SSp	18159	77	24	5.24	2.87
MOs	13347	72	24	3.78	1.91
CP	11696	61	23	2.75	1.07
CA1	10413	85	24	5.54	3.13
MOp	8941	59	23	4.04	1.94
VIsp	8894	59	24	4.63	2.90
DG	7031	80	24	5.26	2.61
CA3	5853	75	24	5.19	2.10
RSPv	4211	41	19	6.49	4.36
RSPd	4082	53	20	5.87	3.52
LSr	4052	49	21	3.35	2.19
RSPagl	3716	47	20	4.16	2.33
LP	3551	39	19	8.56	7.63
ACAAd	3089	52	22	4.58	2.60
SUB	2974	33	16	8.19	5.69
ACAv	2744	36	17	5.21	3.03
VISI	2596	25	14	4.39	2.59
VISal	2541	26	18	4.23	2.50
LGd	2391	35	20	9.40	8.20
PL	2327	19	11	3.46	1.75
PO	2290	26	15	9.13	7.94
APN	2208	26	14	9.37	4.40
VPM	2179	29	16	10.88	8.61
RT	2063	38	20	10.15	6.24
LD	1991	29	16	11.34	8.60
LSc	1920	33	17	1.66	0.85
ILA	1849	18	11	3.14	1.36
MB	1846	46	20	7.91	3.15
VISrl	1840	19	14	4.43	2.34
VISpm	1501	18	11	4.04	2.57
SCig	1453	26	14	5.79	1.66
MGv	1377	23	15	8.76	7.77
LGv	1377	31	15	8.47	4.49
STR	1266	39	21	4.15	1.86
MRN	1255	29	17	7.85	2.94
SCiw	1183	19	11	7.71	2.55
ProS	1045	19	12	5.42	3.41
VISam	1016	11	5	3.76	2.17
TH	1003	48	23	7.88	5.45

Region label	Units	Files	Subjects	firing_rate mean (Hz)	firing_rate median (Hz)
DP	939	15	8	4.03	1.74
VPL	820	21	15	11.55	8.57
VISa	807	10	9	4.30	2.52
SCsg	775	19	10	5.04	3.24
POST	763	17	11	6.22	3.68
MGd	729	21	13	8.19	7.23
No Area	681	66	23	4.64	3.25
ORBvl	678	4	2	5.24	3.62
MGm	632	19	11	9.36	8.77
SF	628	24	14	3.40	2.11
VL	600	43	22	2.91	1.51
TTd	588	10	7	4.51	2.29
CA2	574	37	19	5.72	2.53
MD	572	17	11	7.92	6.25
SGN	562	12	8	7.57	6.21
LSv	508	17	12	3.71	2.44
SSs	502	3	2	4.34	2.13
VISli	444	6	4	4.60	2.97
TRS	433	13	9	4.83	3.36
OLF	414	10	6	4.78	1.95
SCop	410	22	12	4.37	1.78
ORBm	382	6	3	4.55	2.17
CL	376	11	8	7.34	5.70
VAL	371	10	8	10.51	8.36
AV	367	16	12	8.95	7.21
NOT	345	17	10	10.76	4.93
POL	331	10	8	5.82	4.49
PoT	304	11	8	7.62	4.55
PPT	289	9	6	10.12	5.26
Eth	243	11	11	8.02	6.04
IGL	226	13	8	8.61	4.81
ACB	213	3	3	3.77	2.05
PF	204	5	4	5.91	4.26
AD	188	10	8	5.63	3.35
ZI	183	8	6	8.44	5.04
PIL	176	7	6	7.32	3.59
LH	163	7	7	11.67	7.15
MH	157	6	5	5.26	4.32
PVT	131	8	7	4.61	3.19
NPC	129	5	4	9.71	3.02
out of brain	127	16	10	3.30	2.00
HPF	126	19	12	5.50	2.91
IntG	121	4	4	6.77	3.83

Region label	Units	Files	Subjects	firing_rate mean (Hz)	firing_rate median (Hz)
FRP	118	4	3	3.11	1.77
OP	113	5	5	9.13	4.41
TEa	112	1	1	3.10	1.55
PAG	108	3	2	5.70	2.17
MS	103	4	4	3.00	2.07
PT	102	7	4	4.32	3.48
SCdg	100	4	2	6.43	2.67
V3	99	11	10	2.45	1.27
PAR	98	1	1	6.80	5.28
BST	97	6	5	3.17	1.69
SFO	87	2	2	2.25	1.39
VISpor	86	2	1	4.35	2.09
ENTm	80	1	1	7.17	5.58
RPF	71	3	2	11.77	4.96
SCzo	69	12	10	4.48	3.92
NB	68	2	2	3.54	1.85
PRC	68	2	2	7.29	5.45
SEZ	67	14	8	4.31	1.61
IAD	62	8	6	6.38	3.48
AON	62	1	1	3.57	0.87
PP	60	4	4	8.32	5.32
AMd	55	4	4	9.03	5.61
FF	52	1	1	6.42	1.56
GPe	52	1	1	20.46	8.93
IG	46	1	1	2.12	0.58
NLL	38	1	1	6.27	6.44
SPFp	37	3	2	8.64	6.43
PAL	35	4	3	5.77	3.00
ORBI	34	1	1	5.39	2.98
LT	29	2	2	10.77	8.66
SPA	26	2	2	2.49	1.65
PCN	25	1	1	4.26	2.52
SCdw	22	2	1	12.19	2.19
DT	20	4	3	2.30	1.08
PRE	18	4	3	2.68	0.95
AMv	17	1	1	10.06	6.97
MPT	16	2	2	1.00	0.00
SubG	14	3	3	5.47	2.72
CM	14	1	1	8.44	7.38
IMD	12	1	1	8.63	6.56
P	11	1	1	4.18	4.56
FC	4	1	1	2.45	2.14
HY	3	1	1	7.77	2.34

6. Intervals tables (stimuli, trials, epochs)

Each NWB intervals table, with the number of files containing it, total rows, and its columns as stored.

Table	Files	Rows	Columns
dynamic_routing_image_set_presentations	36	187,187	start_time, stop_time, active, duration, end_frame, flashes_since_change, image_name, is_change, is_image_novel, omitted, rewarded, start_frame, stimulus_block, stimulus_name
flash_250ms_presentations	99	14,850	start_time, stop_time, active, color, contrast, duration, end_frame, is_change, is_image_novel, omitted, rewarded, start_frame, stimulus_block, stimulus_index, stimulus_name
gabor_20_deg_250ms_presentations	99	360,855	start_time, stop_time, active, contrast, duration, end_frame, is_change, is_image_novel, omitted, orientation, position_x, position_y, rewarded, spatial_frequency, start_frame, stimulus_block, stimulus_index, stimulus_name, temporal_frequency
image-set-im005_r_presentations	17	88,421	start_time, stop_time, active, duration, end_frame, flashes_since_change, image_name, is_change, is_image_novel, omitted, rewarded, start_frame, stimulus_block, stimulus_name
image-set-im087_r_presentations	14	72,772	start_time, stop_time, active, duration, end_frame, flashes_since_change, image_name, is_change, is_image_novel, omitted, rewarded, start_frame, stimulus_block, stimulus_name
image-set-im104_r_presentations	16	83,194	start_time, stop_time, active, duration, end_frame, flashes_since_change, image_name, is_change, is_image_novel, omitted, rewarded, start_frame, stimulus_block, stimulus_name
image-set-im114_r_presentations	16	83,200	start_time, stop_time, active, duration, end_frame, flashes_since_change, image_name, is_change, is_image_novel, omitted, rewarded, start_frame, stimulus_block, stimulus_name
spontaneous_presentations	99	198	start_time, stop_time, active, duration, end_frame, is_change, is_image_novel, omitted, rewarded, start_frame, stimulus_block, stimulus_name
trials	99	66,130	start_time, stop_time, initial_image_name, change_image_name, is_change, change_time, go, catch, response_time, response_latency, reward_time, reward_volume, hit, false_alarm, miss, correct_reject, aborted, auto_rewarded, change_frame, trial_length, no_reward_epoch, omitted_reward, is_sham_change

7. Column and signal descriptions from the NWB files

Quoted verbatim from sub-604914_ses-20220427T041046.nwb; all files in the Dandiset share this structure.

Location in NWB file	Name	Description (as written in the NWB file)
electrodes	(table)	metadata about extracellular electrodes
electrodes	location	the location of channel within the subject e.g. brain region
electrodes	group	a reference to the ElectrodeGroup this electrode is a part of
electrodes	group_name	the name of the ElectrodeGroup this electrode is a part of
electrodes	probe_vertical_position	Length-wise position of electrode/channel on device (microns)
electrodes	probe_horizontal_position	Width-wise position of electrode/channel on device (microns)
electrodes	probe_id	The unique id of this electrode's/channel's device
electrodes	probe_channel_number	The local index of electrode/channel on device
electrodes	valid_data	Whether data from this electrode/channel is usable
electrodes	x	the x coordinate of the position (+x is posterior)
electrodes	y	the y coordinate of the position (+y is inferior)
electrodes	z	the z coordinate of the position (+z is right)
electrodes	imp	the impedance of the electrode, in ohms
electrodes	filtering	description of hardware filtering, including the filter name and frequency cutoffs
units	(table)	
units	PT_ratio	no description
units	amplitude	no description
units	amplitude_cutoff	no description
units	cluster_id	no description
units	cumulative_drift	no description
units	d_prime	no description
units	firing_rate	no description
units	isi_label	no description
units	isi_violations	no description
units	isolation_distance	no description
units	l_ratio	no description
units	max_drift	no description
units	nn_hit_rate	no description
units	nn_miss_rate	no description
units	peak_channel_id	no description
units	presence_ratio	no description
units	quality	no description
units	recovery_slope	no description
units	repolarization_slope	no description
units	silhouette_score	no description
units	snr	no description
units	spread	no description
units	structure_layer	no description
units	velocity_above	no description
units	velocity_below	no description

Location in NWB file	Name	Description (as written in the NWB file)
units	waveform_duration	no description
units	spike_times	times (s) of detected spiking events
units	spike_amplitudes	amplitude (s) of detected spiking events
units	waveform_mean	mean waveforms on peak channels (over samples)
intervals/dynamic_routing_image_set_p representations	(table)	Presentation times and stimuli details for 'dynamic_routing_image_set' stimuli. Note: image_name references control_description in stimulus/templates
intervals/dynamic_routing_image_set_p representations	start_time	Start time of epoch, in seconds
intervals/dynamic_routing_image_set_p representations	stop_time	Stop time of epoch, in seconds
intervals/dynamic_routing_image_set_p representations	active	No description
intervals/dynamic_routing_image_set_p representations	duration	No description
intervals/dynamic_routing_image_set_p representations	end_frame	No description
intervals/dynamic_routing_image_set_p representations	flashes_since_change	No description
intervals/dynamic_routing_image_set_p representations	image_name	No description
intervals/dynamic_routing_image_set_p representations	is_change	No description
intervals/dynamic_routing_image_set_p representations	is_image_novel	No description
intervals/dynamic_routing_image_set_p representations	omitted	No description
intervals/dynamic_routing_image_set_p representations	rewarded	No description
intervals/dynamic_routing_image_set_p representations	start_frame	No description
intervals/dynamic_routing_image_set_p representations	stimulus_block	Index of contiguous presentations of one stimulus type
intervals/dynamic_routing_image_set_p representations	stimulus_name	Name of stimulus
intervals/dynamic_routing_image_set_p representations	tags	user-defined tags
intervals/dynamic_routing_image_set_p representations	timeseries	index into a TimeSeries object
intervals/flash_250ms_presentations	(table)	Presentation times and stimuli details for 'C:/ProgramData/StimulusFiles/dev/flash_250ms' stimuli. Note: image_name references control_description in stimulus/templates
intervals/flash_250ms_presentations	start_time	Start time of epoch, in seconds
intervals/flash_250ms_presentations	stop_time	Stop time of epoch, in seconds
intervals/flash_250ms_presentations	active	No description
intervals/flash_250ms_presentations	color	No description
intervals/flash_250ms_presentations	contrast	Contrast of stimulus
intervals/flash_250ms_presentations	duration	No description
intervals/flash_250ms_presentations	end_frame	No description
intervals/flash_250ms_presentations	is_change	No description

Location in NWB file	Name	Description (as written in the NWB file)
intervals/flash_250ms_presentations	is_image_novel	No description
intervals/flash_250ms_presentations	omitted	No description
intervals/flash_250ms_presentations	rewarded	No description
intervals/flash_250ms_presentations	start_frame	No description
intervals/flash_250ms_presentations	stimulus_block	Index of contiguous presentations of one stimulus type
intervals/flash_250ms_presentations	stimulus_name	Name of stimulus
intervals/flash_250ms_presentations	tags	user-defined tags
intervals/flash_250ms_presentations	timeseries	index into a TimeSeries object
intervals/gabor_20_deg_250ms_presentations	(table)	Presentation times and stimuli details for 'C:/ProgramData/StimulusFiles/dev/gabor_20_deg_250ms' stimuli. Note: image_name references control_description in stimulus/templates
intervals/gabor_20_deg_250ms_presentations	start_time	Start time of epoch, in seconds
intervals/gabor_20_deg_250ms_presentations	stop_time	Stop time of epoch, in seconds
intervals/gabor_20_deg_250ms_presentations	active	No description
intervals/gabor_20_deg_250ms_presentations	contrast	Contrast of stimulus
intervals/gabor_20_deg_250ms_presentations	duration	No description
intervals/gabor_20_deg_250ms_presentations	end_frame	No description
intervals/gabor_20_deg_250ms_presentations	is_change	No description
intervals/gabor_20_deg_250ms_presentations	is_image_novel	No description
intervals/gabor_20_deg_250ms_presentations	omitted	No description
intervals/gabor_20_deg_250ms_presentations	orientation	Orientation of stimulus
intervals/gabor_20_deg_250ms_presentations	position_x	No description
intervals/gabor_20_deg_250ms_presentations	position_y	No description
intervals/gabor_20_deg_250ms_presentations	rewarded	No description
intervals/gabor_20_deg_250ms_presentations	spatial_frequency	Spatial frequency of stimulus
intervals/gabor_20_deg_250ms_presentations	start_frame	No description
intervals/gabor_20_deg_250ms_presentations	stimulus_block	Index of contiguous presentations of one stimulus type
intervals/gabor_20_deg_250ms_presentations	stimulus_name	Name of stimulus
intervals/gabor_20_deg_250ms_presentations	temporal_frequency	Temporal frequency of stimulus
intervals/gabor_20_deg_250ms_presentations	tags	user-defined tags

Location in NWB file	Name	Description (as written in the NWB file)
intervals/gabor_20_deg_250ms_presentations	timeseries	index into a TimeSeries object
intervals/spontaneous_presentations	(table)	Presentation times and stimuli details for 'spontaneous' stimuli. Note: image_name references control_description in stimulus/templates
intervals/spontaneous_presentations	start_time	Start time of epoch, in seconds
intervals/spontaneous_presentations	stop_time	Stop time of epoch, in seconds
intervals/spontaneous_presentations	active	No description
intervals/spontaneous_presentations	duration	No description
intervals/spontaneous_presentations	end_frame	No description
intervals/spontaneous_presentations	is_change	No description
intervals/spontaneous_presentations	is_image_novel	No description
intervals/spontaneous_presentations	omitted	No description
intervals/spontaneous_presentations	rewarded	No description
intervals/spontaneous_presentations	start_frame	No description
intervals/spontaneous_presentations	stimulus_block	Index of contiguous presentations of one stimulus type
intervals/spontaneous_presentations	stimulus_name	Name of stimulus
intervals/spontaneous_presentations	tags	user-defined tags
intervals/spontaneous_presentations	timeseries	index into a TimeSeries object
intervals/trials	(table)	experimental trials
intervals/trials	start_time	Start time of epoch, in seconds
intervals/trials	stop_time	Stop time of epoch, in seconds
intervals/trials	initial_image_name	NOT IMPLEMENTED: initial_image_name
intervals/trials	change_image_name	NOT IMPLEMENTED: change_image_name
intervals/trials	is_change	NOT IMPLEMENTED: is_change
intervals/trials	change_time	NOT IMPLEMENTED: change_time
intervals/trials	go	NOT IMPLEMENTED: go
intervals/trials	catch	NOT IMPLEMENTED: catch
intervals/trials	lick_times	NOT IMPLEMENTED: lick_times
intervals/trials	response_time	NOT IMPLEMENTED: response_time
intervals/trials	response_latency	NOT IMPLEMENTED: response_latency
intervals/trials	reward_time	NOT IMPLEMENTED: reward_time
intervals/trials	reward_volume	NOT IMPLEMENTED: reward_volume
intervals/trials	hit	NOT IMPLEMENTED: hit
intervals/trials	false_alarm	NOT IMPLEMENTED: false_alarm
intervals/trials	miss	NOT IMPLEMENTED: miss
intervals/trials	correct_reject	NOT IMPLEMENTED: correct_reject
intervals/trials	aborted	NOT IMPLEMENTED: aborted
intervals/trials	auto_rewarded	NOT IMPLEMENTED: auto_rewarded
intervals/trials	change_frame	NOT IMPLEMENTED: change_frame
intervals/trials	trial_length	NOT IMPLEMENTED: trial_length
intervals/trials	no_reward_epoch	NOT IMPLEMENTED: no_reward_epoch
intervals/trials	omitted_reward	NOT IMPLEMENTED: omitted_reward
intervals/trials	is_sham_change	NOT IMPLEMENTED: is_sham_change
acquisition	EyeTracking	

Location in NWB file	Name	Description (as written in the NWB file)
acquisition	v_in	The theoretical maximum voltage that the running wheel encoder will reach prior to "wrapping". This should theoretically be 5V (after crossing 5V goes to 0V, or vice versa). In practice the encoder does not always reach this value before wrapping, which can cause transient spikes in speed at the voltage "wraps".
acquisition	v_sig	Voltage signal from the running wheel encoder
processing/eye_tracking_rig_metadata	(module)	Eye tracking rig metadata module
processing/eye_tracking_rig_metadata	eye_tracking_rig_metadata	
processing/licking	(module)	Licking behavior processing module
processing/licking	licks	Timestamps and stimulus presentation frame indices for lick events
processing/optotagging	(module)	optogenetic stimulation data
processing/optotagging	optotagging	no description
processing/optotagging/optogenetic_stimulation	(table)	
processing/optotagging/optogenetic_stimulation	start_time	Start time of epoch, in seconds
processing/optotagging/optogenetic_stimulation	condition	no description
processing/optotagging/optogenetic_stimulation	level	no description
processing/optotagging/optogenetic_stimulation	stop_time	Stop time of epoch, in seconds
processing/optotagging/optogenetic_stimulation	stimulus_name	no description
processing/optotagging/optogenetic_stimulation	duration	no description
processing/optotagging/optogenetic_stimulation	tags	user-defined tags
processing/optotagging/optogenetic_stimulation	timeseries	index into a TimeSeries object
processing/rewards	(module)	Licking behavior processing module
processing/rewards	autorewarded	no description
processing/rewards	volume	no description
processing/running	(module)	Running speed processing module
processing/running	dx	Running wheel angular change, computed during data collection
processing/running	speed	no description
processing/running	speed_unfiltered	no description
processing/stimulus	(module)	Stimulus Times processing
processing/stimulus	timestamps	no description

8. Exported files

File	Rows	Columns
tables/sessions.csv	99	session_id, asset_path, size_gb, subject_id, species, sex, age, genotype, strain, session_start_time, session_description, institution, lab, n_units, n_electrodes, electrode_groups, region_source, regions, intervals, acquisition, processing
tables/units.parquet	201,192	session_id, unit_id, unit_row, region, PT_ratio, amplitude, amplitude_cutoff, cluster_id, cumulative_drift, d_prime, firing_rate, isi_label, isi_violations, isolation_distance, l_ratio, max_drift, nn_hit_rate, nn_miss_rate, peak_channel_id, presence_ratio, quality, recovery_slope, repolarization_slope, silhouette_score, snr, spread, structure_layer, velocity_above, velocity_below, waveform_duration, subject_id
tables/region_summary.csv	126	region, n_units, n_sessions, n_subjects, firing_rate_mean_hz, firing_rate_median_hz
tables/intervals_summary.csv	9	table, n_sessions, n_rows, columns
intervals/<table>.parquet	one per intervals table	session_id + the table's columns
tables/dataset_metadata.json, nwb_file_metadata.json, nwb_descriptions.csv		published metadata quoted in sections 1 and 7